

MATH 130

Test 3 Review

Trigonometry and Vectors

90-Minute Review Session



Session Roadmap

0–15 min

Compound Interest & Logarithms

15–30 min

Solving Exponential & Log Equations

30–50 min

Angle Conversions (DMS, Radians)

50–65 min

Arc Length & Sector Area

65–80 min

Triangle & Trapezoid Area

80–90 min

Parallel Lines, Circles & Linear Speed

Compound Interest Formulas

Discrete Compounding

$$A = P(1 + r/n)^{(nt)}$$

P = principal (initial amount)

r = annual interest rate (decimal)

n = compounds per year

t = time in years

Continuous Compounding

$$A = Pe^{(rt)}$$

e $\approx$ 2.71828...

Same P, r, t as discrete formula

Used when interest compounds "continuously"



Common n values: monthly = 12, daily = 365, quarterly = 4, weekly = 52

Example: P = \$2500, r = 3%, t = 20 years

n	12 (monthly)	365 (daily)	Continuous
A	\$4,551.89	\$4,555.18	\$4,555.30

Worked Example: Compound Interest

\$2500 at 3% for 20 years, compounded monthly ($n = 12$) Continuous compounding:

Step 1: Identify: $P=2500$, $r=0.03$, $n=12$,
 $t=20$

Step 2: $A = 2500(1 + 0.03/12)^{(12 \cdot 20)}$

Step 3: $A = 2500(1.0025)^{(240)}$

Step 4: $A = 2500(1.82075) = \$4,551.89$

Step 1: $A = Pe^{(rt)} = 2500 \cdot e^{(0.03 \cdot 20)}$

Step 2: $A = 2500 \cdot e^{(0.6)}$

Step 3: $A = 2500 \cdot 1.82212 = \$4,555.30$



Calculator tip: Use the y^x or $^$ button. Enter $(1.0025)^{240}$ carefully — parentheses matter!

Try It!

Compound Interest

1. \$3,000 invested at 5% for 15 years compounded quarterly. Find A.
2. \$8,000 at 2.5% for 10 years compounded daily. Find A.
3. \$1,500 at 4% for 6 years compounded continuously. Find A.
4. How much more does continuous earn vs. monthly for \$5,000 at 3% over 25 years?



Answers

1. \$6,321.54
2. \$10,272.12
3. \$1,906.87
4. \$9.90 more

Exponential $\leftrightarrow$ Logarithmic Form

$$b^y = x \iff \log_b(x) = y$$

base^{exponent} = result $\leftrightarrow$ log_{base}(result) = exponent

Exp $\rightarrow$ Log

$$25^{(5/2)} = 3125$$

$$\rightarrow \log_{25}(3125) = 5/2$$

Log $\rightarrow$ Exp

$$\log_4(2048) = 5.5$$

$$\rightarrow 4^{5.5} = 2048$$



Memory trick: The base stays the base! The log answer is the exponent.
Think: "The log asks: what power do I raise the base to, to get this number?"

Solving Exponential & Log Equations

Example A: $\ln x = 4$

$\ln x = 4$ means $\log_e(x) = 4$

Convert to exponential form:

$$x = e^4$$

$$x = 54.598$$

Example B: $2^x = 13$

Take $\ln$ (or $\log$) of both sides:

$$\ln(2^x) = \ln(13)$$

$$x \cdot \ln(2) = \ln(13)$$

$$x = \ln(13) / \ln(2)$$

$$x \approx 3.700$$



Key Log Properties

$$\ln(a^n) = n \cdot \ln(a)$$

|

$$\ln(e) = 1$$

|

$$e^{(\ln x)} = x$$

|

$$\log(10) =$$

Try It!

1. Write $8^{2/3} = 4$ in logarithmic form.

2. Write $\log_5(625) = 4$ in exponential form.

3. Solve: $\ln x = 7$ (round to 3 decimal places)

4. Solve: $5^x = 40$ (round to 3 decimal places)



Answers

1. $\log_8(4) = 2/3$

2. $5^4 = 625$

3. $x = e^7 \approx 1096.633$

4. $x = \ln 40 / \ln 5 \approx 2.292$

Angle Conversions: DMS $\leftrightarrow$ Decimal

DMS $\rightarrow$ Decimal

$$D^{\circ} M' S'' \rightarrow D + M/60 + S/3600$$

Example: $37^{\circ}41'11''$

$$\begin{aligned} &= 37 + 41/60 + 11/3600 \\ &= 37 + 0.6833 + 0.00306 \\ &= \mathbf{37.6864^{\circ}} \end{aligned}$$

Decimal $\rightarrow$ DMS

Whole part = Degrees

Example: 36.3492°

$$D = 36^{\circ}$$

$$0.3492 \times 60 = 20.952 \rightarrow M = 20'$$

$$0.952 \times 60 = 57.12 \rightarrow S \approx 57''$$

$$= \mathbf{36^{\circ}20'57''}$$



Remember: $1^{\circ} = 60$ minutes, 1 minute = 60 seconds. Always take only the decimal part for the next step!

Radians $\leftrightarrow$ Degrees

$$\pi \text{ radians} = 180^\circ$$

Radians $\rightarrow$ Degrees

Multiply by $180/\pi$

$$\begin{aligned} 2.7 \text{ rad} &\times (180/\pi) \\ &= 2.7 \times 57.2958^\circ \\ &\approx \mathbf{154.7^\circ} \end{aligned}$$

Degrees $\rightarrow$ Radians

Multiply by $\pi/180$

$$\begin{aligned} 163.9^\circ &\times (\pi/180) \\ &= 163.9 \times 0.01745 \\ &\approx \mathbf{2.86 \text{ rad}} \end{aligned}$$

Common Values:

$30^\circ = \pi/6$

$45^\circ = \pi/4$

$60^\circ = \pi/3$

$90^\circ = \pi/2$

$180^\circ = \pi$

$360^\circ = 2\pi$

Try It!

Angle Conversions

1. Convert $52^{\circ}15'30''$ to decimal degrees (4 places).

2. Convert 128.4375° to DMS.

3. Convert 3.5 radians to degrees (1 decimal).

4. Convert 225° to radians (2 decimal places).



Answers

1. 52.2583°

2. $128^{\circ}26'15''$

3. 200.5°

4. 3.93 rad

Arc Length

$$s = r \cdot \theta \quad (\theta \text{ must be in RADIANS!})$$

a) $r = 5.7, \quad \theta = 65^\circ$

First convert 65° to radians:

$$\theta = 65 \times \pi/180 = 1.1345 \text{ rad}$$

$$s = 5.7 \times 1.1345$$

$$s \approx 6.5$$

b) $r = 10, \quad \theta = 3\pi/5$

θ is already in radians!

$$s = 10 \times (3\pi/5)$$

$$s = 30\pi/5 = 6\pi$$

$$s \approx 18.8$$

⚠ #1 Mistake: Forgetting to convert degrees to radians before using $s = r\theta$!

Sector Area

$$A = \frac{1}{2} \cdot r^2 \cdot \theta \quad (\theta \text{ must be in RADIANS!})$$

a) $r = 5.7, \quad \theta = 65^\circ$

Convert: $\theta = 65 \times \pi/180 = 1.1345 \text{ rad}$

$$A = \frac{1}{2} \times 5.7^2 \times 1.1345$$

$$A = \frac{1}{2} \times 32.49 \times 1.1345$$

$$A \approx 18.4$$

b) $r = 10, \quad \theta = 3\pi/5$

Already in radians!

$$A = \frac{1}{2} \times 10^2 \times (3\pi/5)$$

$$A = \frac{1}{2} \times 100 \times 1.8850$$

$$A \approx 94.2$$

Arc length: $s = r\theta$ (linear)

vs.

Sector area: $A = \frac{1}{2}r^2\theta$ (squared)

Try It!

Arc Length & Sector Area

1. Find arc length: $r = 8$, $\theta = 72^\circ$

2. Find arc length: $r = 15$, $\theta = 2\pi/3$

3. Find sector area: $r = 8$, $\theta = 72^\circ$

4. Find sector area: $r = 15$, $\theta = 2\pi/3$



Answers

1. $s \approx 10.1$

2. $s \approx 31.4$

3. $A \approx 40.2$

4. $A \approx 235.6$

Triangle Area: Heron's Formula

$$s = (a + b + c) / 2$$
$$A = \sqrt{[s(s-a)(s-b)(s-c)]}$$

Example: $a = 1.5 \text{ cm}$, $b = 1.3 \text{ cm}$, $c = 2.1 \text{ cm}$

$$s = (1.5 + 1.3 + 2.1) / 2 = 4.9 / 2 = 2.45$$

$$s - a = 0.95, \quad s - b = 1.15, \quad s - c = 0.35$$

$$A = \sqrt{(2.45 \times 0.95 \times 1.15 \times 0.35)}$$

$$A = \sqrt{0.93734} \approx 1.0 \text{ cm}^2$$

More Area Formulas

Triangle: $A = \frac{1}{2}bh$

Use when base & height are known.

Test example: $b = 775\text{m}$, $h = 123\text{m}$

$$A = \frac{1}{2} \times 775 \times 123$$

$$A = 47,662.5 \text{ m}^2$$

Trapezoid: $A = \frac{1}{2}(b_1+b_2)h$

b_1 & b_2 = parallel sides, h = height

Test: $b_1=2.05$, $b_2=6.25 \text{ cm}$

Find h using Pythagorean thm

$$h \approx 1.96 \text{ cm}$$

$$A = \frac{1}{2}(2.05+6.25)(1.96) \approx 8.13 \text{ cm}^2$$



When to use which formula:

$A = \frac{1}{2}bh$ → you know base and perpendicular height

Heron's → you know all 3 sides but NOT the height

Trapezoid → two parallel bases + height (may need Pythagorean thm)

Worked Example: Multi-Step Trapezoid

Step 1: Find the Height (Pythagorean Thm)

Given: $b_1 = 4.5 \text{ m}$, $b_2 = 10.1 \text{ m}$, $\text{slant} = 7.0 \text{ m}$

$$d = 10.1 - 4.5 = 5.6 \text{ m}$$

$$h^2 + 5.6^2 = 7.0^2 \rightarrow h^2 = 49 - 31.36 = 17.64$$

$$h = \sqrt{17.64} = 4.2 \text{ m}$$

Step 2: Compute the Area

$$A = \frac{1}{2}(b_1 + b_2)h$$

$$A = \frac{1}{2}(4.5 + 10.1)(4.2)$$

$$A = \frac{1}{2}(14.6)(4.2)$$

$$A = 30.66 \text{ m}^2$$



Key strategy for trapezoid problems:

Step 1: Find the horizontal difference $d = b_2 - b_1$. This is one leg of the hidden right triangle.

Step 2: Use Pythagorean theorem: $h^2 + d^2 = \text{slant}^2$, solve for h .

Step 3: Plug into $A = \frac{1}{2}(b_1 + b_2)h$. Don't forget square units!

Try It!

Triangle & Trapezoid Area

1. Heron's: $a = 5$ cm, $b = 7$ cm, $c = 9$ cm. Find area.
2. Triangle: base = 18 m, height = 11 m. Find area.
3. Trapezoid: bases 8 ft & 14 ft, height 6 ft. Find area.
4. Heron's: $a = 3.2$ m, $b = 4.1$ m, $c = 5.6$ m. Find area.



Answers

1. $A \approx 17.4 \text{ cm}^2$
2. $A = 99.0 \text{ m}^2$
3. $A = 66.0 \text{ ft}^2$
4. $A \approx 6.5 \text{ m}^2$

Parallel Lines & Proportional Segments

If parallel lines cut transversals, corresponding segments are proportional.

$$AB / BD = AC / CE \quad (\text{corresponding segments on different transversals})$$

a) Find CE

$$AB = 40, BD = 45, AC = 37$$

$$40/45 = 37/CE$$

$$CE = 37 \times 45/40$$

$$CE = 41.6 \text{ m}$$

b) Find DC

DC is part of the transversal:

$$DC = CE - AC$$

$$DC = 41.6 - 37$$

$$DC = 4.6 \text{ m}$$

Circle Theorems: Arcs & Angles

Inscribed Angle

= $\frac{1}{2}$ (intercepted arc)

Central Angle

= intercepted arc

Intersecting chords

$\angle = \frac{1}{2}(\text{arc}_1 + \text{arc}_2)$ for vertical angles at intersection

Semicircle

AC is diameter $\rightarrow$ arc AC = 180°

Worked: $\angle AED = 133^\circ$, arc AB = 52° , AC is diameter

a) $\angle AEB$

Linear pair with $\angle AED$
 $180^\circ - 133^\circ$
= 47°

b) $\angle ACB$

Inscribed $\angle$, arc AB = 52°
 $\frac{1}{2} \times 52^\circ$
= 26°

c) arc CD

arc BC = 128° , intersecting
 $133 = \frac{1}{2}(AD + 128)$
AD = 138° , CD = $180 - 138$
= 42°

Linear & Angular Speed

Angular speed: $\omega = \theta/t$

(radians per unit time)

Linear speed: $v = r\omega$

(distance per unit time)

1 revolution = 2π radians

Example: 350 RPM, blade = 3 ft

Step 1: Find angular speed

$$\omega = 350 \text{ rev/min} \times 2\pi \text{ rad/rev} = 700\pi \text{ rad/min}$$

Step 2: Find linear speed

$$v = r\omega = 3 \times 700\pi = 2100\pi \approx 6,597.3 \text{ ft/min}$$

Step 3: Convert to mph

$$6597.3 \text{ ft/min} \times 60 \text{ min/hr} \div 5280 \text{ ft/mi} \approx 75.0 \text{ mph}$$

Try It!

1. Inscribed angle intercepts a 96° arc. What is the angle?
2. Two chords intersect. One pair of vertical angles makes arcs of 80° and 50° . Find the angle.
3. A wheel spins at 500 RPM with radius 2 ft. Find ft/min.
4. Convert 12,566.4 ft/min to mph.



Answers

1. 48°
2. $\frac{1}{2}(80+50) = 65^\circ$
3. $v = 2000\pi \approx 6283.2$
4. ≈ 142.8 mph

? Practice Quiz

10 problems — try them all before checking answers!

1. \$4,000 at 6% for 10 years compounded monthly. Find A.

2. Write $\log_3(81) = 4$ in exponential form.

3. Solve: $e^x = 20$ (3 decimal places).

4. Convert $45^\circ 22' 18''$ to decimal degrees (4 places).

5. Convert 128.75° to DMS (nearest second).

6. Convert $5\pi/6$ radians to degrees (exact).

7. Arc length: $r = 12$, $\theta = 40^\circ$.

8. Sector area: $r = 7$, $\theta = 2\pi/3$.

9. Heron's formula: $a = 5$, $b = 7$, $c = 9$. Find area.

10. 600 RPM, radius 1.5 ft. Find linear speed in mph.



Practice Quiz – Answers

1. \$4,000 at 6%, $n=12$, $t=10$

\$7,277.59

2. $\log_3(81) = 4 \rightarrow$ exponential

$3^4 = 81$

3. $e^x = 20 \rightarrow x = \ln(20)$

$x \approx 2.996$

4. $45^\circ 22' 18'' \rightarrow$ decimal

45.3717°

5. $128.75^\circ \rightarrow$ DMS

$128^\circ 45' 00''$

6. $5\pi/6 \rightarrow$ degrees

150°

7. Arc: $r=12$, $\theta=40^\circ$

$s \approx 8.4$

8. Sector: $r=7$, $\theta=2\pi/3$

$A \approx 51.3$

9. Heron's: 5, 7, 9

$A \approx 17.4$

10. 600 RPM, $r=1.5$ ft $\rightarrow$ mph

≈ 64.3 mph

Formula Cheat Sheet

Compound Interest

$$A = P(1 + r/n)^{(nt)}$$

Continuous

$$A = Pe^{(rt)}$$

Log $\leftrightarrow$ Exp

$$b^y = x \Leftrightarrow \log_b(x) = y$$

DMS $\rightarrow$ Decimal

$$D + M/60 + S/3600$$

Rad $\rightarrow$ Deg

$$\times 180/\pi$$

Deg $\rightarrow$ Rad

$$\times \pi/180$$

Arc Length

$$s = r\theta \quad (\text{rad})$$

Sector Area

$$A = \frac{1}{2}r^2\theta \quad (\text{rad})$$

Heron's

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

Trapezoid

$$A = \frac{1}{2}(b_1 + b_2)h$$

Inscribed $\angle$

$$= \frac{1}{2} (\text{intercepted arc})$$

Linear Speed

$$v = r\omega$$

Good luck on the test! Remember: show all work for full credit.